

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Final Examination of Semester 1

Year: 2024

Subject: 031001123 Chemistry III

Section: 15-18

Date: 4 October 2024

Time: 10.00-12.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. The test is divided into 3 sections. There will be 7 pages (including this page), 8 questions and they are worth 90 marks (40 points).

- This exam paper contains no errors. If a suspected error is found, it is the student's discretion to correct it.
- Answer the questions on this test paper.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators cannot be used in this test.

This exam paper is part of a set used to assess student abilities as follows:

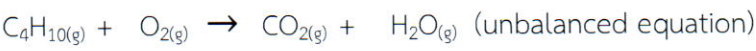
		Question (Q)
CLO 1	Explanation and comparison between covalent bond, ionic bond, and metallic bond.	
CLO 2	Description of theoretical term and calculation of chemical bond energies.	Q3, Q7, Q8
CLO 3	Application of fundamental chemistry in other subjects.	Q1, Q2, Q4, Q5, Q6
CLO 4	Realization of discipline, punctuality, and public and self-responsibility.	

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

Part1 : Total score 30 marks

1. Consider the following unbalanced chemical equation:



1.1) What is the sum of all coefficients of the reactants and products in the balanced equation? (2 Marks)

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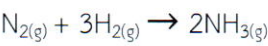
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1.2) Write the rate expressions for the disappearance of butane (C_4H_{10}) and molecular oxygen, and the formation of carbon dioxide and water. (4 Marks)

Ans. Rate of reaction = _____ = _____ = _____ = _____

1.3) For the following balanced equation:



Write the rate expressions for the disappearance of nitrogen ($\Delta[N_2]/\Delta t$) in terms of the formation of ammonia (NH_3). (4 Marks)

Ans. $\frac{\Delta[N_2]}{\Delta t} =$ _____

2. The reaction $2A_{(aq)} \rightarrow B_{(aq)}$. If the initial concentration of A is 0.5 M, and after 20 minutes of the reaction, the concentration of A becomes 0.1 M:

2.1) Calculate the average rate of the reaction over the period from 0 to 20 minutes. (6 Marks)

Ans. Average rate of the reaction = _____

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2.2) What is the concentration of B at 20 minutes? (6 Marks)

Ans. [B] at 20 minutes = _____ M

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3. Predict (mark ✓) how the following factors affect the rate of the decomposition of N_2O_5 reaction: (8 Marks)



Parameter	Increase	Decrease	None
Decreased temperature			
Increased concentration of N_2O_5			
Presence of a catalyst			
Adding N_2 gas			

Part 2 : Total score 30 marks

4. At 25 °C, the following data were collected for the reaction $2A + B \rightarrow C$ (20 marks)

Experiment	[A] (M)	[B] (M)	Rate (M/s)
1	0.10	0.10	2.0×10^{-4}
2	0.20	0.10	8.0×10^{-4}
3	0.10	0.20	4.0×10^{-4}
4	0.01	0.01	?

- 4.1) Determine the rate law for the reaction.
- 4.2) Calculate the rate constant, k .
- 4.3) Calculate the rate when the $[A]$ and $[B] = 0.01 \text{ M}$.
- 4.4) What is the overall order of the reaction?
- 4.5) Suppose the reaction with a rate law: $\text{Rate} = k[A][B]^2$. If the concentration of A is doubled and the concentration of B is reduced to one-half, Calculate the rate of reaction changes by those factors.

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5.1) Determine the rate constant (k) of the reaction.

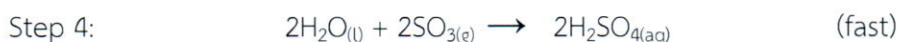
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Part 3 : Total score 30 marks

6. Consider the following mechanism of sulfuric acid (H_2SO_4)

(10 marks)



6.1) Write the overall reaction equation?

6.2) What are intermediates?

6.3) Identify any catalysts in the reaction? Which step added the catalysts to the reaction?

6.4) Determine rate of reaction?

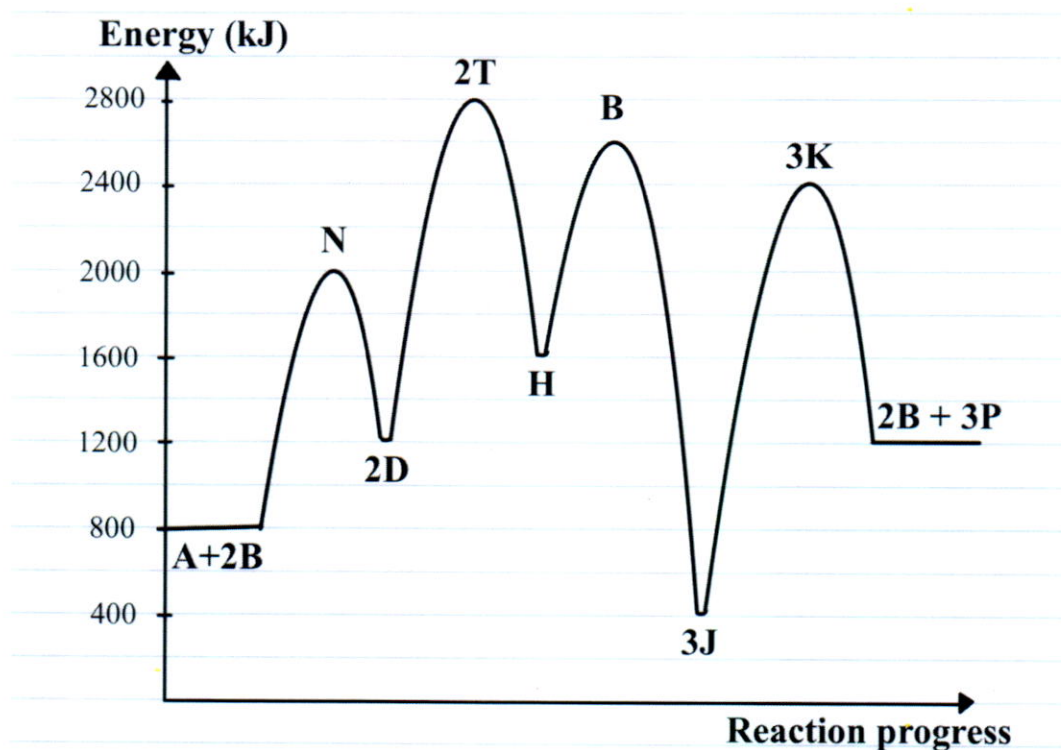
6.5) Which is the rate-determining step of this reaction? Please explain your reason.

7. What is the meaning of catalyst? Why are catalysts used in industry?

(5 marks)

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8. Using the data illustrated in the graphs below to answer the following equations. (15 marks)



8.1) How many elementary steps are there? Write all the mechanisms of this reaction.

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8.2) Write the overall reaction equation?

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8.3) What are intermediates?

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8.4) Which is the rate-determining step? Please explain your reason.

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8.5) Calculate E_a for each reaction? What is the E_a for the overall reaction?

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1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18		
1 H Hydrogen 1.008	Atomic # Symbol Name Weight																		2 He Helium 4.0026
3 Li Lithium 6.94	4 Be Beryllium 9.0122	C Solid Hg Liquid H Gas Rf Unknown																9 F Fluorine 18.998	10 Ne Neon 20.180
11 Na Sodium 22.990	12 Mg Magnesium 24.305	Alkali metals Alkaline earth metals Transition metals Post-transition metals Metalloids Other nonmetals Noble gases																17 Cl Chlorine 35.45	18 Ar Argon 39.948
19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.630	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798		
37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29		
55 Cs Caesium 132.91	56 Ba Barium 137.33	57-71 89-103																86 Rn Radon (222)	
7 Fr Francium (223)	88 Ra Radium (226)	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)			
		104 Rf Rutherfordium (267)	105 Db Dubnium (268)	106 Sg Seaborgium (269)	107 Bh Bohrium (270)	108 Hs Hassium (277)	109 Mt Meitnerium (278)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (290)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)			
For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.																			
57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97					
89 Ac Actinium (227)	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)									

$$e^{-1} = 0.36$$

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Examiners